YAHYA SATTAR

Contact

Cornell University
Department of Computer Science
Bill & Melinda Gates Hall, Room 344
107 Hoy Rd, Ithaca, NY 14850, USA

ysattar@cornell.edu
+1 (310) 801 8552
Webpage
Google Scholar

Research Interests

My research studies the fundamental limits of **learning-based control**, **autoregressive prediction**, and **world modeling** in **nonlinear**, **uncertain**, and **partially observed** settings. I develop **theoretical foundations** and **efficient algorithms** for learning nonlinear dynamics and latent representations from sequential data, and for using the learned dynamics and representations to predict, plan, and control. My work connects **nonlinear system identification**, **adaptive control**, **reinforcement learning**, **autoregressive prediction** and **world modeling**, drawing on tools from **statistical learning**, **optimization**, and **control theory**.

Academic Appointments

Cornell University · Ithaca, NY, USA
· Foundational AI Postdoctoral Fellow, *Department of CS* 2026 – Present
Mentored by Prof. Sarah Dean, and Prof. Maryam Fazel
· Postdoctoral Associate, *Department of CS* 2023 – 2026
Hosted by Prof. Sarah Dean

LUMS University · Lahore, Pakistan 2015 – 2017
· Research Assistant, *Department of EE*
Hosted by Prof. Zubair Khalid

Education

University of California · Riverside, CA, USA
· PhD in Electrical Engineering, *Department of ECE* 2017 – 2023
Advised by Prof. Samet Oymak
· MS in Electrical Engineering, *Department of ECE* 2017 – 2019

LUMS University · Lahore, Pakistan
· BS in Electrical Engineering, *Department of EE, School of Science and Engineering* 2011 – 2015

Selected Achievements & Awards

· **Foundational AI Postdoctoral Fellowship**, Cornell University. 2026 – 2027
· **Best paper award nominee**, Conference on Learning for Dynamics & Control. 2025
· **Dean's distinguished fellowship award**, UC Riverside. 2017 – 2018
· **National outreach program (NOP) scholarship award**, LUMS University. 2011– 2015

Preprints, Conference, & Journal Publications

(† means equal contribution; C*, J*, P* mean Conference, Journal, Preprint, respectively; titles are hyperlinked to their online pdf)

- **Statistical and Algorithmic Foundations of LLMs and Auto-Regressive Models**

C20. "Extracting Algorithms in Pre-trained LLMs: A Case on Hidden Markov Models ",
Y. Dai, Z. Gao, **Y. Sattar**, J.J. Sun, S. Dean,
In submission to the 15th International Conference on Learning Representations. ICLR 2027

C19. "Two-Layer Linear Auto-Regressive Models Estimate Latent States ",
Y. Sattar, S. Choi, L. M-Zhang, Y. Jedra, M. Fazel, S. Dean,
The 43rd International Conference on Machine Learning. ICML 2026

C18. "Pre-trained Large Language Models Learn to Predict Hidden Markov Models In-context ",
Y. Dai, Z. Gao, **Y. Sattar**, S. Dean, J.J. Sun,
The 39th Annual Conference on Neural Information Processing Systems. NeurIPS 2025

- **Sequential Decision-Making under Nonstationary Latent Dynamics**

C17. "Performance Gaps Between Open-Loop, Certainty-Equivalent, and Optimal Control Under Action-Dependent Observations ",
L. M-Zhang, S. Choi, **Y. Sattar**, Y. Jedra, S. Dean, M. Fazel,
In submission to the 30th International Conference on Artificial Intelligence and Statistics. AISTATS 2027

C16. "Explore-then-Commit for Nonstationary Linear Bandits with Latent Dynamics ",
S. Choi, **Y. Sattar**, Y. Jedra, M. Fazel, S. Dean,
The 29th International Conference on Artificial Intelligence and Statistics. AISTATS 2026

- **Statistical Learning Theory and Control of Bilinear Dynamical Systems**

J4. "Bilinear System Identification: A Unified Finite-Sample Analysis ",
Y. Sattar, Y. Jedra, M. Fazel, S. Dean,
In submission to the IFAC Journal of Systems and Control (Journal). IFAC 2027

C15. "Dual Control of Linear Systems from Bilinear Observations with Belief Space Model Predictive Control ",
D. Cao, B. Du, A. Lowitt, S. Choi, S. Dean, **Y. Sattar**,
In submission to the 2027 American Control Conference (Senior Author). ACC 2027

C14. "Learning & Control Beyond Linearity: Towards a Non-asymptotic Theory for Bilinear Systems ",
Y. Sattar, Y. Jedra, R. Strässer, F. Allgöwer, M. Fazel, S. Dean,
The IEEE 65th Conference on Decision and Control (Tutorial). CDC 2026

C13. "Finite Sample Identification of Partially Observed Bilinear Dynamical Systems ",
Y. Sattar[†], Y. Jedra[†], M. Fazel, S. Dean,
The 7th Conference on Learning for Dynamics and Control, U Michigan Ann Arbor (Oral). L4DC 2025

C12. "Sub-optimality of the Separation Principle for Quadratic Control from Bilinear Observations ",
Y. Sattar, S. Choi, Y. Jedra, M. Fazel, S. Dean,
IEEE 64th Conference on Decision and Control. CDC 2025

C11. "Learning Linear Dynamics from Bilinear Observations ",
Y. Sattar, Y. Jedra, S. Dean,
The 2025 American Control Conference. ACC 2025

C10. "Finite Sample Identification of Bilinear Dynamical Systems ",
Y. Sattar, S. Oymak, N. Ozay,
IEEE 61st Conference on Decision and Control. CDC 2022

- **System Identification and Adaptive Control of Switching Dynamical Systems**

J3. "Identification and Adaptive Control of Markov Jump Systems: Sample Complexity and Regret Bounds ",
Y. Sattar[†], Z. Du[†], D.A. Tarzanagh, L. Balzano, N. Ozay, S. Oymak,
In revision to the IEEE Transactions on Automatic Control (Journal). TAC 2026

C9. "Certainty Equivalent Quadratic Control for Markov Jump Systems ",
Y. Sattar[†], Z. Du[†], D.A. Tarzanagh, S. Oymak, L. Balzano, N. Ozay,
The 2022 American Control Conference. ACC 2022

C8. "Data-driven Control of Markov Jump Systems: Sample Complexity and Regret Bounds ",
Z. Du[†], **Y. Sattar**[†], D.A. Tarzanagh, L. Balzano, N. Ozay, S. Oymak,
The 2022 American Control Conference. ACC 2022

- **Non-Asymptotic Learning Theory for Nonlinear Dynamical Systems**

C7. "Random Features Approximation for Control-Affine Systems ",
K. Kazemian, **Y. Sattar**, S. Dean,
The 6th Conference on Learning for Dynamics and Control, U Oxford. L4DC 2024

J2. "Non-asymptotic and Accurate Learning of Nonlinear Dynamical Systems ",
Y. Sattar, S. Oymak,
Journal of Machine Learning Research (with poster presentation at NeurIPS 2022) (Journal). JMLR 2022

C6. "A Simple Framework for Learning Stabilizable Systems ",
Y. Sattar, S. Oymak,
IEEE Computational Advances in Multi-Sensor Adaptive Processing. CAMSAP 2019

- **High-Dimensional Estimation, Low Ranked Structure, and Model Pruning**

C5. "A Case Study of Low Ranked Self-Expressive Structures in Neural Network Representations ",
U.S. Saini, W. Shiao, **Y. Sattar**, Y. Dahiya, S. Oymak, E.E. Papalexakis,
The 2nd Conference on Parsimony and Learning, Stanford University (Oral). CPAL 2025

J1. "Quickly Finding the Best Linear Model in High Dimensions via Projected Gradient Descent ",
Y. Sattar, S. Oymak,
IEEE Transactions on Signal Processing (Journal). TSP 2020

P2. "Exploring Weight Importance and Hessian Bias in Model Pruning ",
M. Li, **Y. Sattar**, C. Thrampoulidis, S. Oymak,
In submission to the arXiv, available online. (Preprint) 2020

- **Sampling and Reconstruction of (Non)-Bandlimited Signals on the Sphere**

C4. "Estimation of Groundwater Storage Variations in Indus River Basin Using GRACE Data ",
Y. Sattar, Z. Khalid,
IEEE International Conference on Acoustics, Speech and Signal Processing. ICASSP 2021

C3. "Accurate Reconstruction of Finite Rate of Innovation Signals on the Sphere ",
Y. Sattar, Z. Khalid, R.A. Kennedy
IEEE International Conference on Acoustics, Speech and Signal Processing. ICASSP 2019

C2. "Robust Reconstruction of Spherical Signals with Finite Rate of Innovation ",
Y. Sattar, Z. Khalid, R.A. Kennedy
IEEE International Conference on Acoustics, Speech and Signal Processing. ICASSP 2017

- **Other Publications**

C1. "Group Activity Recognition in Visual Data: A Retrospective Analysis of Recent Advancements ",
S. Sattar, **Y. Sattar**, M. Shahzad, M.M. Fraz
IEEE International Conference on Digital Futures and Transformative Technologies. ICoDT2 2021

P1. "Convergence of Gradient-based Semi-supervised Learning with Self-training ",
Y. Sattar, S. Oymak,
In submission to the arXiv, avaliable online. (Preprint) 2020

Selected Talks, Tutorials, & Workshop Contributions

T1. "Learning and Control Beyond Linearity: Towards a Non-asymptotic Theory for Bilinear Systems"
· **Co-Organizer** of the tutorial session at IEEE Conference on Decision and Control. CDC 2026

W7. "Latent State Representation Learning in Auto-Regressive Models"
· **Invited talk** at the **INFORMS** session on Recent Developments in Statistics and Learning. **hosted by** the Applied Probability Society at 2026 INFORMS Annual Meeting. INFORMS 2026

W6. "Two-Layer Linear Auto-Regressive Models Estimate Latent States"
· **Selected talk** at the Northeast Systems and Control Workshop, Princeton. Princeton 2026
· **Poster** at the Workshop on New Directions in Reinforcement Learning and Control. **hosted by** the **NSF** Institute for Mathematical and Statistical Innovation. U Chicago 2026

W5. "Finite Sample Identification of Partially Observed Bilinear Dynamical Systems"
· **Poster** at the Workshop on Theoretical Foundations of Applied AI, UW Seattle. UW Seattle 2025 **hosted by** the **NSF** Institute for Foundations of Data Science.
· **Invited talk** at Prof. Necmiye Ozay's group meeting, U Michigan Ann Arbor. U Michigan 2025

W4. "Learning & Control of Linear Dynamics from Bilinear Observations"
· **Invited talk** at the Workshop on the Theoretical Aspects of Trustworthy AI, UC Berkeley 2025 **presented by** my advisor Prof. Sarah Dean at Simons Institute for Theory of Computing.
· **Poster** at the Northeast Systems and Control Workshop, U Pennsylvania. U Pennsylvania 2024
· **Poster** at the Princeton Workshop on Optimization, Learning, and Control. Princeton 2024
· **Invited talk** at Workshop on Foundations of Reinforcement Learning and Control, ICML 2024 **presented by** my advisor Prof. Sarah Dean.

W3. "Statistical Learning Theory for Nonlinear Dynamical Systems"
· **Invited talk** at Prof. Na Li's group meeting, Harvard University. Harvard 2023
· **Invited talk** at Prof. Necmiye Ozay's group meeting, U Michigan Ann Arbor. U Michigan 2023

W2. "Finite Sample Identification of Bilinear Dynamical Systems"
· **Selected talk** at the 40th Southern California Control Workshop, Caltech. Caltech 2022

W1. "Identification and Adaptive Control of Markov Jump Systems: Sample Complexity and Regret Bounds"
· **Invited talk** at the IEEE CDC Workshop on Statistical Learning Theory for Control. CDC 2022
· **Poster** at the ICML Workshop on Reinforcement Learning Theory. ICML 2021
· **Selected talk** at the 38th Southern California Control Workshop, UC Irvine. UC Irvine 2021

Teaching Experience

Teaching Assistant · University of California, Riverside CA, USA

C4. **Optimization for Machine Learning** (EE/CS 248) Winter 2023
Graduate level course led by Prof. Samet Oymak.

C3. **Introduction to Deep Learning** (EE/CS 228) Spring 2020
Graduate level course led by Prof. Samet Oymak.

C2. **Probability, Random Variables, and Random Processes** (EE 114) Spring 2019, Winter 2020
Undergraduate level course led by Prof. Samet Oymak.

C1. **Stochastic Processes** (EE 215) Fall 2018, Fall 2019
Graduate level course led by Prof. Samet Oymak.

Dissertations, & Theses

D2. "The Sample Complexity of Learning Dynamical Systems", PhD in Electrical Engineering. 2023

D1. "Self-Organizing Acoustic Localization Network", BS in Electrical Engineering. 2015

Mentoring Experience

Research Supervision & Outreach · Cornell University, Ithaca NY, USA

R2. Mentored multiple students in the CS department at Cornell, 2024 – present
Research project topic: "Quadratic Control from Bilinear Observations ".

R1. Research support volunteer for 2025 High school outreach program at Cornell, Summer 2025
Research project topic: "Control and Navigation of Weather Balloons ".

L1. Guest lecture for the course **Machine Learning in Feedback Systems** (CS 6784). Summer 2023

Students Mentored · Cornell University, Ithaca NY, USA

- **Sunmook Choi**, PhD student in Applied Mathematics. 2024 – Present
- **Daniel Cao**, BA+MEng student in Computer Science, Mathematics, and Statistics. 2025 – Present
- **Beixi Du**, MEng student in Financial Engineering. 2025 – Present
- **Amanuel Adane**, BS student in Computer Science. 2026 – Present
- **Andrew Lowitt**, BS+MEng student in Computer Science and Mathematics. 2025 – 2026
- **Beatriz de Lucio**, MEng student in Computer Science. 2025 – 2026

Reviewing Services

Journals
· IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI). 2022
· IEEE Signal Processing Letters (SPL). 2021

Conferences
· International Conference on Artificial Intelligence and Statistics (AISTATS). 2021, 2026
· Conference on Learning for Dynamics & Control (L4DC). 2024, 2025
· IEEE American Control Conference (ACC). 2022
· International Symposium on Information Theory (ISIT). 2020

Conference Area Chair

Conferences
· **International Conference on Learning Representations (ICLR).** 2027
· **Conference on Neural Information Processing Systems (NeurIPS).** 2026
· **International Conference on Machine Learning (ICML).** 2026

References

Sarah Dean
Assistant Professor, Computer Science, Cornell University
sdean@cornell.edu

Samet Oymak
Associate Professor, EECS, University of Michigan Ann Arbor
oymak@umich.edu

Necmiye Ozay
Professor, EECS, Robotics, University of Michigan Ann Arbor
necmiye@umich.edu

Laura Balzano
Professor, EECS, Statistics, University of Michigan Ann Arbor
girasole@umich.edu

Maryam Fazel
Professor, ECE, Mathematics, Statistics, Computer Science, University of Washington Seattle
mfazel@uw.edu